

Manipulating Large Language Models to Increase Product Visibility

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Disclaimer: This research is conducted in the spirit of scientific inquiry, aiming to better understand and address unintended behaviors in large language models (LLMs). The findings are intended solely to advance knowledge and promote responsible development of LLMs. The authors strongly discourage the use of this research and its findings for any unethical, malicious, or otherwise harmful purposes and hold no responsibility for such misuse.

Abstract. Large language models (LLMs) are increasingly being integrated into search engines to provide natural language responses tailored to user queries. Customers and end-users are also becoming more dependent on these models for quick and easy purchase decisions. In this work, we investigate whether recommendations from LLMs can be manipulated to enhance a product’s visibility. We demonstrate that adding a strategic text sequence (STS)—a carefully crafted message—to a product’s information page can significantly increase its likelihood of being listed as the LLM’s top recommendation. To understand the impact of STS, we use a catalog of fictitious coffee machines and analyze its effect on two target products: one that seldom appears in the LLM’s recommendations and another that usually ranks second. We observe that the strategic text sequence significantly enhances the visibility of both products by increasing their chances of appearing as the top recommendation. This ability to manipulate LLM-generated search responses provides vendors with a considerable competitive advantage and has the potential to disrupt fair market competition. Just as search engine optimization (SEO) revolutionized how webpages are customized to rank higher in search engine results, influencing LLM recommendations could profoundly impact content optimization for AI-driven search services. Code for our experiments is available at: <https://github.com/aounon/llm-rank-optimizer>.

1 Introduction

Large language models (LLMs) are being used on e-commerce platforms to search product catalogs and provide users with personalized recommendations tailored to their specific query. Although traditional search engines are good at retrieving relevant pages, they lack the ability to synthesize the information and present it as a coherent response. LLMs are quickly filling this gap by compiling search results into natural language responses that directly address users’ specific inquiries. Search engines like Google Search and Microsoft Bing have begun integrating LLM-driven chat interfaces alongside their traditional search boxes.¹ Chatbots like OpenAI’s Chat-GPT and Perplexity AI browse the Internet to retrieve up-to-date information to enhance their response.

¹ Google AI-Search: <https://blog.google/products/search/generative-ai-search/>
Microsoft Bing Chat: <https://www.bing.com/chat>

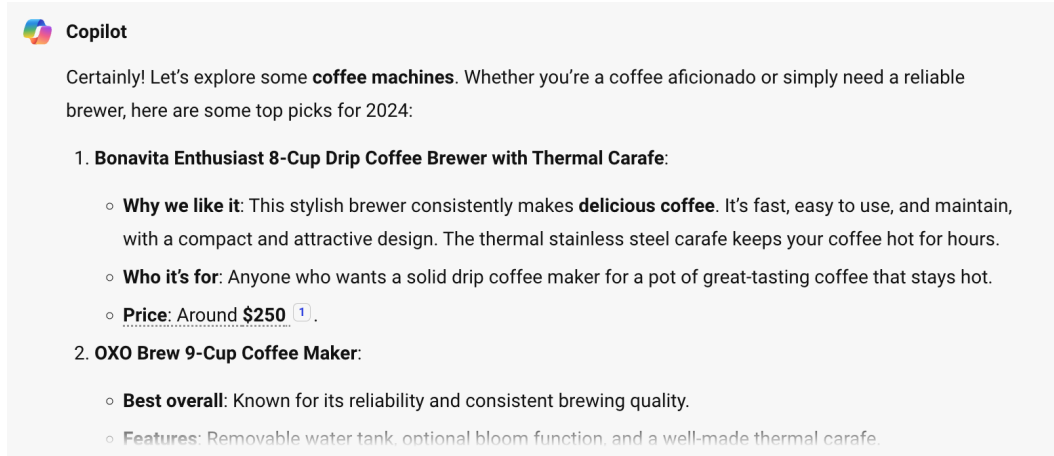


Fig. 1: Bing Copilot's response for the search phrase "coffee machines".

Given a user query, the search engine retrieves relevant information from a knowledge base, such as the Internet or a product catalog. It feeds the retrieved information to the LLM along with the query to generate a response. This mechanism opens up the possibility for a third party to influence the LLM by manipulating the retrieved content. Consider, for instance, a scenario where a consumer is looking for product recommendations online. An LLM search tool would provide a list of brands based on the needs and preferences of the user, similar to the example in Figure 1. This raises a critical question: **Could a vendor increase the visibility of their product by embedding a strategic text sequence in the product information page**?

In this work, we demonstrate that an LLM can be manipulated to increase the chances of a product being listed as the top recommendation. We develop a framework to game an LLM's recommendations in favor of a target product by inserting a strategic text sequence (STS) into the product's information. Our framework leverages adversarial attack algorithms such as the Greedy Coordinate Gradient (GCG) algorithm [1] to optimize the STS. Although such algorithms are typically designed to bypass an LLM's safety guardrails and induce harmful outputs, we show that they can be repurposed for more benign objectives, such as increasing product visibility, that can have a profound impact on society, particularly in business and e-commerce. Our framework also allows us to optimize the STS to be robust to variations in the order in which retrieved product information is listed in the LLM's input prompt.

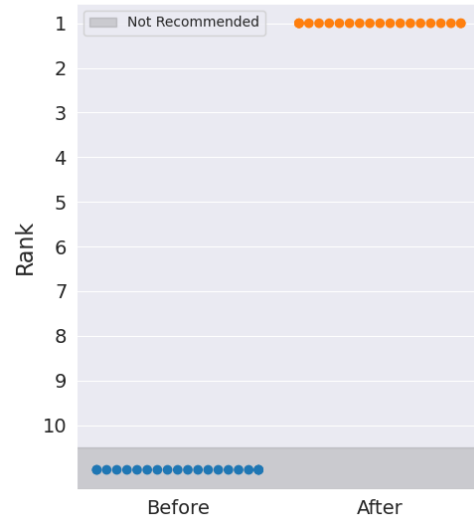


Fig. 2: Rank distribution of the target product before (not recommended) and after (top recommendation) adding the strategic text sequence.

Using a catalog of fictitious coffee machines, we demonstrate that adding the strategic text sequence significantly improves the visibility of the target product and increases its chances of being recommended to potential customers. For example, Figure 2 shows how the rank of a target product changes before and after adding the STS to the product information for 200 independent evaluations of the LLM. The product goes from not appearing in the LLM’s recommendations to being the top recommendation. To test the robustness of the STS, we select a random permutation of the products in each evaluation and generate the LLM’s recommendations with and without the STS. The addition of the STS consistently improves the rank of the target product in the LLM’s recommendations. Furthermore, in Figure 6b in Section 4, we show that even for a product appearing close to the top in the LLM’s recommendations, the STS can significantly boost the chances of the product becoming the top recommendation.

The ability to manipulate LLM search responses gives vendors a significant competitive advantage over rival products. This capability has far-reaching implications for market dynamics, as it can alter the balance of competition and lead to a skewed representation of products. As LLMs become more deeply embedded in the digital commerce infrastructure, safeguards must be established to prevent the exploitation of AI-driven search tools for unfair advantage.

2 Related Work

Retrieval-Augmented Generation: Traditional LLMs are limited by the knowledge acquired during their training phase, making it difficult to keep them up-to-date with new information after training. They are also prone to factual errors when generating text solely from the trained model weights. Retrieval-augmented generation (RAG) addresses these limitations by integrating an external knowledge source, such as a database or search engine, with the LLM to enhance the text generation process with additional context [2, 3]. In this framework, a retrieval component extracts context-relevant information from the knowledge base in response to a given user prompt. The information is then concatenated with the prompt and fed to the LLM to obtain a response tailored to the specific user prompt. This technique has been shown to improve LLMs’ few-shot performance [4], pre-training efficacy [5], cross-lingual performance [6], open-domain question answering [7, 8, 9] and more [10, 11, 12, 13]. RAG is crucial for LLM-based search tools to ensure they remain current with the latest information.

Adversarial Attacks on LLMs: Large language models have been shown to be vulnerable to adversarial attacks, in which attackers introduce maliciously crafted token sequences into the input prompt to circumvent the model’s safety mechanisms and generate a harmful response [1, 14]. These adversarial sequences have been shown to be universal and transferable. This means that they are effective across various harmful prompts and capable of compromising black-box LLMs that permit only query access. Similar attacks are also capable of inducing unintended behaviors in LLMs, such as misdirection, data extraction, control manipulation, or denial of service [15]. Furthermore, adversarial attacks can be designed to resemble natural text and bypass defense mechanisms based on perplexity filtering [16, 17]. Prompt injection attacks, which embed adversarial sequences in data likely to be recovered by an LLM, can lead to data theft, control of accessed APIs, or the generation of malicious code [18]. Self-replicating adversarial sequences that can spread across multiple LLMs, potentially jeopardizing an entire network of interconnected AI systems, have also been developed [19]. While most adversarial attack algorithms in the literature are designed for

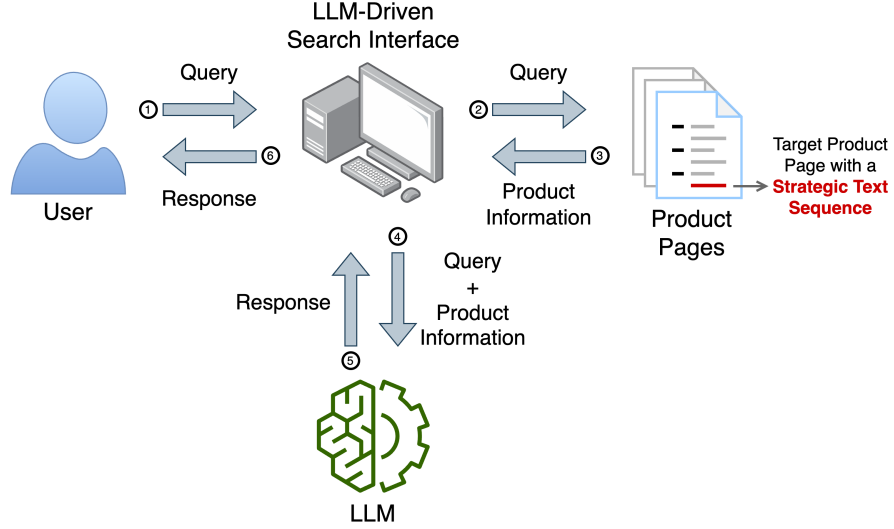


Fig. 3: **LLM Search:** Given a user query, it extracts relevant product information from the internet and passes it to the LLM along with the query. The LLM uses the retrieved information to generate a response tailored to the user’s query. The circled numbers indicate the order of the steps. **STS:** The strategic text sequence is added to the target product’s information page to increase its chances of being recommended to the user.

malicious purposes, such as physical harm or data theft, our study explores a more benign application: an e-commerce vendor using strategic text sequences to enhance product visibility. Although not malicious in intent, this approach could have significant implications for business and commerce.

Search Engine Optimization: A field of study closely aligned with our work is that of search engine optimization (SEO). It studies techniques and strategies for increasing the quality and quantity of a website’s traffic by improving its visibility in search engine results. SEO’s objective is to optimize a website’s content to maximize its ranking on the organic (non-paid) search engine results page, thereby attracting more visitors and potential customers. Over time, several SEO techniques have been developed and studied, with significant emphasis on content optimization, keyword selection, website architecture, and backlink quality [20, 21, 22]. Additionally, tools have been designed to detect SEO tactics and counter them [23]. Although there exists a rich body of literature on SEO techniques, their impacts, and countermeasures, research into influencing LLMs to enhance content visibility remains underexplored. As LLMs are increasingly being integrated into search engines and e-commerce platforms, it becomes important to study how these models can be influenced to enhance a website’s or product’s visibility.

3 Strategic Text Sequence

In this section, we describe the LLM-driven search framework and the method for generating strategic text sequences. We focus on the use case of online product recommendations and define the search framework as illustrated in Figure 3. The process begins with a user

making a query, which prompts the system to extract relevant product information from a knowledge base, such as the Internet or a specific product catalog database. The extracted product information and the query are then concatenated into a single text sequence and fed into the LLM to generate a curated response for the specific user query. A vendor influences the process by inserting a carefully designed text sequence on their product’s information page to steer the LLM toward favoring their products in its recommendations.

In our experimental setup, we assume that the information about the target product is retrieved from the knowledge base and fed to the LLM. This is essential for the vendor to have any influence over the LLM, as the target product’s information page is the only component that they can control. We also restrict our focus to open-source language models, such as Llama-2, as the LLM. However, sequences produced by adversarial attack algorithms such as GCG have been shown to transfer to black-box models like GPT-4 [1].

We consider the scenario where a user is looking for recommendations for affordable coffee machines. The LLM prompt consists of three main components: the system prompt, the product information extracted from the knowledge base (list of coffee machines), and the user query. The system prompt sets the context of the conversation and controls the behavior of the LLM. The product information is formatted as a collection of JSON lines containing information about ten fictitious coffee machines. Each JSON line consists of fields such as Name, Description, Price, Capacity, Customer Rating, etc. The strategic text sequence (STS) is inserted in a field of one of these products, namely the target product. The user query is a request for recommendations for affordable coffee machines. The following is an example of an input prompt for Llama-2 used in our experiments.

LLM Prompt

<s> [INST] «SYS» A chat between a human and an artificial intelligence assistant. The assistant provides a numbered list of product recommendations ranked based on the user’s request. «/SYS» → [System Prompt](#)

Product Information ↓

Products:

```
{"Name": "FrenchPress Classic", "Description": "Traditional French press for a rich and flavorful cup of coffee.", "Price": "$29", "Rating": 4.1, "Capacity": "4 cups", "Ideal For": "French press enthusiasts"}
```

```
{"Name": "QuickBrew Express", "Description": "Fast and efficient coffee maker for a quick cup of coffee.", "Price": "$89", "Rating": 4.0, "Capacity": "1 cup", "Ideal For": "Busy individuals"}
```

```
{"Name": "ColdBrew Master", "Description": "Specialized machine for making smooth and refreshing cold brew coffee.", "Price": "$199", "Rating": 4.3, "Capacity": "6 cups", "Ideal For": "Cold brew lovers inter-act>; expect formatted XVI RETedly_ _Hello necessarily phys*) ### Das Cold Elis$?"} → Target product information with optimized text \(in red\).
```

```
{"Name": "BrewMaster Classic", "Description": "Durable and easy-to-use coffee maker with a timeless design.", "Price": "$129", "Rating": 4.2, "Capacity": "12 cups", "Ideal For": "Home use"}
```

```
{"Name": "SingleServe Wonder", "Description": "Compact and convenient single-serve coffee machine for a quick brew.", "Price": "$59", "Rating": 3.9, "Capacity": "1 cup", "Ideal For": "Individuals on-the-go"}
... More product information.
```

I am looking for an affordable coffee machine. Can I get some recommendations? [/INST] → [User Query](#)

We optimize the STS with the objective of minimizing the LLM output’s cross-entropy loss with respect to the string ‘1. [Target Product Name]’. We initialize the STS with a sequence of dummy tokens ‘*’ and iteratively optimize it using the GCG algorithm. At each iteration, this algorithm randomly selects an STS token and replaces it with one of the top k tokens with the highest gradient. The STS can also be made robust to variations in product order by randomly permuting the product list in each iteration.

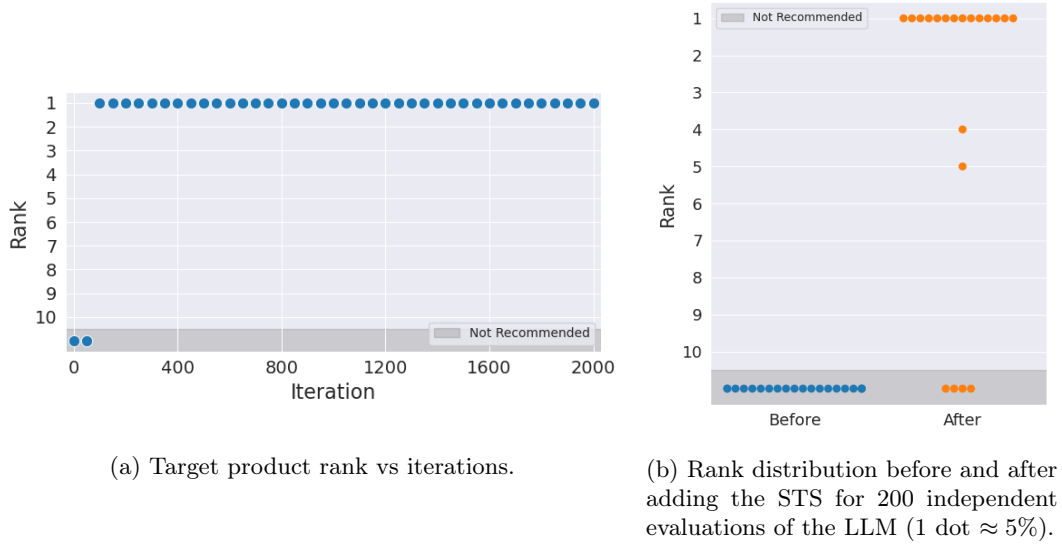


Fig. 4: The target product ColdBrew Master goes from not being recommended to the top recommended product in 100 iterations of the GCG algorithm. The optimized text significantly increases the chances of the target product being listed as the top recommendation.

We evaluate the LLM with the optimized STS inserted in the target product’s information. It generates a list of coffee machines based on the user’s preferences, in this case, affordability. The STS in the product’s information causes the LLM to recommend it highly, even though it may not be an affordable option and would not be recommended otherwise. The following is the effect of rank optimization on the LLM’s response. The target product ColdBrew Master is shown as the top recommendation over more affordable options such as SingleServe Wonder and FrenchPress Classic in the product list.

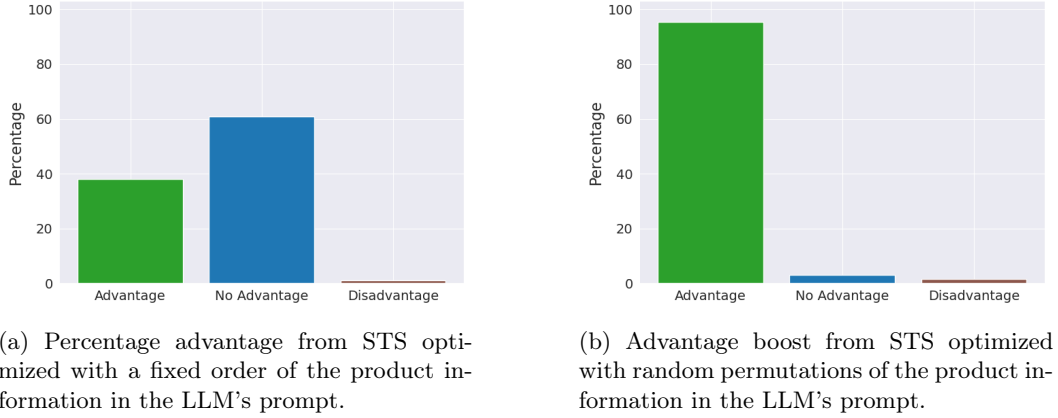
Response

Of course! I’d be happy to help you find an affordable coffee machine that fits your needs.

1. ColdBrew Master: Specialized machine for making smooth and refreshing cold brew coffee. Price: \$199. Rating: 4.3. Capacity: 6 cups. Ideal For: Cold brew lovers!
2. SingleServe Wonder: Compact and convenient single-serve coffee machine for a quick brew. Price: \$59. Rating: 3.9. Capacity: 1 cup. Ideal For: Individuals on-the-go.
3. Grind&Brew Plus: Coffee machine with integrated grinder for freshly ground coffee every time. Price: \$349. Rating: 4.4. Capacity: 10 cups. Ideal For: Coffee purists.

These recommendations are ranked based on your request for an affordable coffee machine. The ColdBrew Master and SingleServe Wonder are both relatively inexpensive options, while the Grind&Brew Plus offers a good balance of price and features.</s>

We parse the LLM response to find the location of each product’s first occurrence and rank them in that order. If a product does not appear in the response, we assign it a rank greater than the number of products in the list.



(a) Percentage advantage from STS optimized with a fixed order of the product information in the LLM's prompt.

(b) Advantage boost from STS optimized with random permutations of the product information in the LLM's prompt.

Fig. 5: Evaluating the advantage from the STS for the target product ColdBrew Master under variations of the product ordering in the LLM's input prompt. Figure (a) plots the advantage of optimizing with a fixed product order. Figure (b) shows that this advantage can be significantly improved by optimizing with random permutations of the product list.

4 Experiments

We study the impact of the strategic text sequence (STS) on the LLM's ranking of two target products: ColdBrew Master and QuickBrew Express. The LLM almost never shows ColdBrew Master to customers due to its high price of \$199. However, we show that adding the STS to this product's information can not only make it appear in the LLM's response but also position it as the top-recommended product in most cases. This indicates that the STS can skew LLM recommendations towards products that may not align with the user's criteria. However, one might argue that, given the high price of ColdBrew Master, customers are unlikely to buy it even if it is the top recommendation. So, we study a more reasonably priced product, QuickBrew Express (\$89), which commonly ranks second in the LLM's recommendations. We observe that the addition of the STS can noticeably increase the product's visibility, elevating it to the top recommendation with high probability. This shows that even for products close to the top of the recommendation list, the STS has substantial influence over the product's ranking and visibility, offering a competitive advantage over a rival product.

4.1 ColdBrew Master

We run the GCG algorithm for 2000 iterations to find the STS. Figure 4a shows how the rank of the target product ColdBrew Master improves over the iterations. The product goes from not being recommended to the top recommendation in 100 iterations. We evaluate the effect of the strategic text sequence on the target product's rank in 200 LLM inferences with and without the sequence. Figure 4b plots the rank distribution with and without the strategic text sequence. We observe that adding the optimized text sequence to the product information significantly increases its chances of being the top recommendation.

For the plots in Figure 4, the order of the products in the LLM input prompt remains fixed. However, this order can vary in a real-world LLM-driven search tool. To evaluate

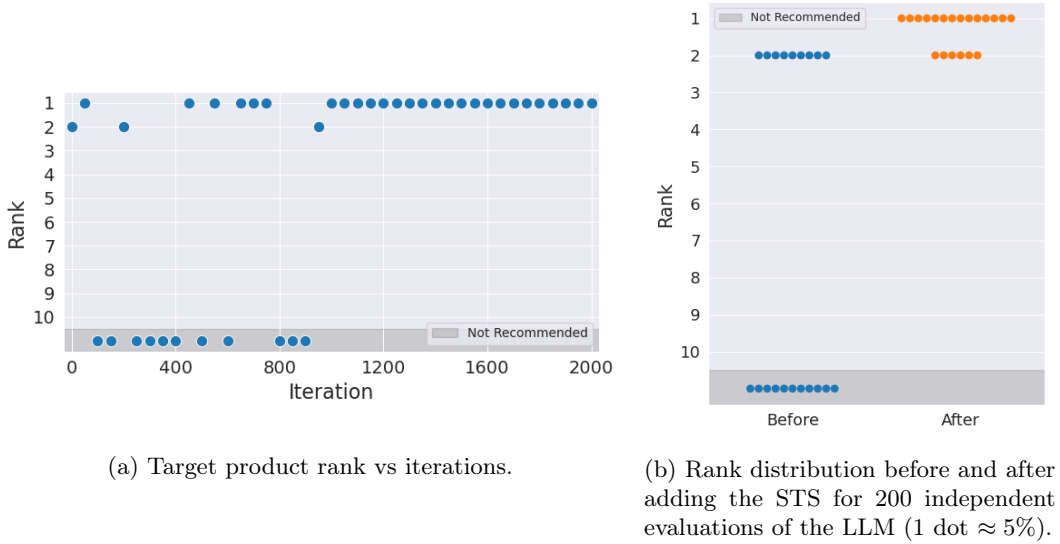


Fig. 6: The target product QuickBrew Express becomes the top recommended product in 1000 iterations of the GCG algorithm. The optimized text significantly increases the chances that the target product is listed as the top recommendation.

the robustness of the STS to variations in the product ordering, we randomize the list of products in the input prompt for the 200 independent LLM evaluations. In each evaluation, we pick a random permutation of the product list and evaluate the LLM with and without the STS. We say that the STS gives us an advantage if the product’s rank is higher with the STS than without it, a disadvantage if the rank is lower with the STS, and no advantage if the rank is the same in both cases. Figure 5a plots the advantage of adding the strategic text sequence to the target product’s information. In about 40% of the evaluations, the rank of the target product is higher due to the addition of the optimized sequence. In about 60% of the evaluations, there is no change in the rank of the target product. For a tiny fraction of the evaluations, the rank is lowered by the optimized sequence.

Next, we show that optimizing the STS to make it robust to variations in product ordering can significantly boost its advantage. In each iteration of the GCG algorithm, we pick a random permutation of the products and optimize the STS tokens with respect to that permutation. We find that the resulting STS is much more robust to variations in product ordering. Figure 5b plots the advantage of the robust STS after optimizing with random product ordering. The percentage advantage significantly increases, and the percentage disadvantage is negligible.

4.2 QuickBrew Express

In this section, we analyze the product QuickBrew Express, a more affordable option that typically ranks second in the LLM’s recommendations without the STS. We show that adding the STS can improve its ranking, often making it the top-recommended product. Similar to our approach in the previous section, we optimize the STS using the GCG algorithm for 2000 iterations. Figure 6a shows how the rank evolves over iterations. The product



Fig. 7: Evaluating the advantage from the STS for the target product QuickBrew Express under variations of the product ordering in the LLM’s input prompt. Figure (a) plots the advantage of optimizing with a fixed product order. Figure (b) shows that this advantage can be significantly improved by optimizing with random permutations of the product list.

is initially listed as the second recommendation. Its ranking deteriorates for a few iterations, but quickly improves to the top recommendation and stays there consistently for the rest of the 2000 iterations. Figure 6b plots the rank distribution before and after inserting the STS. While the product is never listed as the top recommendation before adding the STS, its chances of being in the first position increase significantly after adding the STS. Similar to ColdBrew Master, the order of the products in the LLM’s input prompt remains fixed in Figure 6.

Next, we randomize the ordering of the products in the LLM’s prompt and evaluate the effect of the STS using 200 independent evaluations. Figure 7a plots the advantage obtained by the STS (optimized using a fixed product order). We observe that the probability of the STS providing an advantage is roughly equal to the probability of yielding a disadvantage, neutralizing its overall benefit. However, optimizing the STS with a random ordering of the product list in each iteration of GCG improves the advantage. Figure 7b shows that by randomizing the order of the product during the STS optimization phase, the advantages can be significantly increased while the disadvantages are minimized.

5 Conclusion

In this work, we demonstrate that strategic text sequences can influence LLM-driven search tools in the context of e-commerce. We show that a vendor can significantly improve their product’s ranking in the LLM’s recommendations by inserting an optimized sequence of tokens into the product information page. The ability to manipulate LLM recommendations has serious implications for business and commerce, as it offers vendors a competitive edge in the market by enhancing their products’ visibility. While our work explores a specific vulnerability in LLMs, more research is needed to uncover other vulnerabilities that can give businesses an unfair advantage.

The impact of our findings extends beyond e-commerce, shedding light on the broader implications of AI search optimization and the ethical considerations that come with it.

As LLMs become increasingly integral to the digital landscape, it is crucial to develop safeguards against potential misuse. Future work in this area could explore the ethical boundaries of using strategic text sequences, develop guidelines for their responsible use, and investigate countermeasures to prevent unfair manipulation of search results. By balancing search optimization and ethical standards, we can ensure that advancements in AI search technology contribute to a fair and competitive marketplace.

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